

Project Report

Smart Bed Project

**cw5125 Chutong Wang
yh6548 Ya-cheng Hsieh
yo2282 Ying Ou
rl5721 Rui Liu
kk6146 Kanokkan Kanjanapiban**

**1600_Managing Technical Projects
2025_Fall semester**

Content

Commentary	6
Integration Management (Rui Liu)	7
1. Project Charter	7
2. Develop the Project Management Plan	7
3. Project Execution	8
4. Project Knowledge Management.....	9
5. Monitoring Project Work	10
6. Implement Integrated Change Control.....	10
7. Close Project or Phase	11
Scope Management (Rui Liu)	12
1. Scope planning	12
2. Collect requirements	12
3. Scope definition.....	12
4. WBS.....	17
5. Scope Verification	23
Quality Management (Kanokkan).....	25
1. Quality Objectives	25
2. Quality Assurance (QA)	25
Hardware Testing	26
Firmware Testing.....	27
AI Model Testing.....	27
Cloud & Backend Testing	27
Mobile App Testing.....	27
System Integration Testing	27
User Acceptance Testing (UAT)	27
4. Acceptance Criteria.....	27
Quality Management - Testing (Kanokkan).....	29
1. Project Manager – Teammate A	29
Department: Integration, Schedule, Cost, Risk.....	29
Responsibilities:.....	29
Tools Used:	29
Deliverables:.....	29
2. Product Manager – Teammate A.....	30
Department: Product Requirements, UI/UX, User Stories	30
Responsibilities:.....	30
Tools Used:	30
Deliverables:.....	30

3. System Architect – Teammate B.....	30
Department: End-to-End Architecture (Sensors → Firmware → AI → Cloud → App).....	30
Responsibilities:.....	30
Tools Used:	30
Deliverables:.....	31
4. Hardware Engineering Team – Teammate C.....	31
Department: Sensors, PCB, Electrical Systems	31
Responsibilities:.....	31
Tools Used:	31
Deliverables:.....	31
5. Embedded Firmware Team – Teammate D	31
Department: Firmware Code, Data Synchronization	31
Responsibilities:.....	31
Tools Used:	32
Deliverables:.....	32
6. AI / Algorithm Team – Teammate E	32
Department: Machine Learning, Sleep Analytics	32
Responsibilities:.....	32
Tools Used:	32
Deliverables:.....	32
7. Cloud & Backend Team – Teammate F	33
Department: Cloud Infrastructure, API, Data Storage.....	33
Responsibilities:.....	33
Tools Used:	33
Deliverables:.....	33
8. Mobile App Development Team – Teammate G.....	33
Department: Mobile UI/UX + Bed Control Features.....	33
Responsibilities:.....	33
Tools Used:	33
Deliverables:.....	34
9. Testing & QA Team – Teammate H.....	34
Department: Hardware + Software + System Testing	34
Responsibilities:.....	34
Tools Used:	34
Deliverables:.....	34
10. Procurement, Manufacturing & Operations – Teammate I	34
Department: Supplier Management, Production, Logistics.....	34

Responsibilities:.....	34
Tools Used:	35
Deliverables:.....	35
Stakeholder Management.....	36
Procurement Management	41
Resource Management (Ying Ou)	47
1. Resource Categories.....	47
2. Coordination with Outsourced Vendors	50
3. Resource Requirements by Project Phase	51
Cost Management (Ying Ou)	52
1. Cost Estimate (Kanokkan & Ying Ou).....	52
2. Cost Breakdown by Project Phase	55
3. Cost Control Approach	56
4. Key Cost Risks.....	56
Schedule (Erica)	57
Phase 1: Research & Concept (Month 1–2)	57
Project Proposal & Plan (M1–M2)	57
Phase 2: Product & Technical Design (Month 3–4).....	57
Product Design (M3–M4).....	57
Appearance & Structural Design (M3–M4)	57
Technical Architecture (M3–M4).....	57
Phase 3: Hardware & Software Development (Month 4–8)	57
Hardware Development (M4–7).....	57
Embedded & Algorithm Dev (M4–8)	57
Cloud & Mobile App (M4–8).....	58
Phase 4: Testing, Certification & Pilot Production (Month 8–11).....	58
Product Testing (M8–11)	58
Certification (M10–11)	58
User Acceptance Testing (M10–11)	58
Phase 5: Mass Production & Launch (Month 11–12)	58
Production & Delivery	58
Post-Launch Monitoring (M12+)	59
Communication Management (Erica)	60
Risk Management (Kanokkan)	63
1. Technical Risks.....	63
• Sensor Inaccuracy or Instability.....	63
• Firmware Integration Failures.....	63
• AI Model Accuracy Below Requirements	63
• System Latency or Cloud Data Loss	63

• Insufficient Testing Coverage	63
• Failure to Pass FCC/CE/UL Certification.....	63
• Late Discovery of High-Severity Defects	64
3. Data Security & Privacy Risks.....	64
• Exposure of Sensitive Health Data.....	64
• Cloud Server Downtime Affecting Report Generation	64
4. Operational & Supply Chain Risks	64
• Shortage or Delay of Critical Sensors	64
• Manufacturing Partner Delays.....	64
5. User Experience Risks.....	64
• Mobile App Confusion or Instability	64
• Incorrect or Unreliable Emergency Alerts.....	65
6. Ongoing Risk Monitoring & Control	65
• Continuous Review of Risks Across All Project Phases	65

Commentary

1- how you organized your group project,

We organized our group project by clearly dividing responsibilities based on each team member's strengths and the project management framework. At the beginning of the project, we first reviewed the project requirements and overall objectives together to ensure everyone had the same understanding. After that, we structured the project using a Work Breakdown Structure (WBS), which helped us break the project into manageable sections such as integration, scope, procurement, quality, testing, and risk management.

Each team member was assigned ownership of specific areas. For example, some members focused on system design and development, while others handled management-related sections. This clear role assignment reduced overlap and confusion and allowed each member to work efficiently on their own section.

Moreover, we used regular communication to stay aligned. We texting to the group chat to review progress, discuss issues, and make adjustments when needed. We also used shared documents and project management tools to track tasks and updates. As a result, our group was able to coordinate different technical and management components effectively and keep the project on schedule.

2- how you proceeded in your collaboration efforts

In terms of collaboration, we adopted a sub-teams method and milestone-based workflow. Because our team had more members than most other groups, at the early stage, we divided the work into two smaller sub-teams. This allowed us to stay organized, avoid confusion, and complete tasks more efficiently. In the last few weeks, each member worked on their assigned knowledge area while maintaining regular communication through chatting online and shared documents. We used a shared document to exchange drafts, review each other's work, and ensure consistency in formatting and conceptual alignment. Throughout the process, we frequently cross-checked content. For example, aligning WBS items with schedule phases, and verifying that contents in different knowledge areas did not overlap, so that our final draft reflected a coherent project narrative.

3- how you decided to finalize the approach or focus for the final draft,

4- what were the challenges you faced as a team,

As a team, it was important to make sure everyone understood the main focus and scope of the project. This was challenging because each member had a slightly different interpretation of the objectives. Without aligning these ideas first, the sections we worked on could easily move in different directions.

Another challenge was keeping all sections consistent, since the different management components were connected. A change in one area such as scope management could affect other areas like schedule management and cost management. Because of this, revisions were never limited to just one place. We often had to check the related sections to see if they were affected. To avoid confusion, we later made sure to confirm who was responsible for each part and which sections might be impacted before making changes. This helped reduce mistakes, although it took extra time.

5- what was good about the experience.

Integration Management (Rui Liu)

1. Project Charter

The project charter is jointly developed by the project manager and the project sponsor during the project start-up phase, clarifying the following content and serving as the basis for all subsequent management activities:

Project Vision: To research, develop, and launch a smart bed product with multi-sensor fusion for sleep data collection system and AI analysis, personalized reports, and apnoea alerts.

Overall Milestone Schedule: Complete design, development, testing, certification, and mass production in 12 months.

- Phase 1: Research & Concept (Month 1–2)
- Phase 2: Product & Technical Design (Month 3–4)
- Phase 3: Hardware & Software Development (Month 4–8)
- Phase 4: Testing, Certification & Pilot Production (Month 8–11)
- Phase 5: Mass Production & Launch (Month 11–12)

High-Level Requirements: Reliable sensing system, >85% sleep staging accuracy, APP control, cloud-based sleep reports, passed FCC/CE/UL certification, no high-severity defects in UAT.

Overall Budget: \$4,500,000 - \$9,000,000.

Key Stakeholders: Project Sponsor (Sophie), Project Manager (Peter), Product Manager (Jack), System Architect (Coco), Hardware/Software Team (Candy/David), Algorithm Team, Test Team and QA, Supply Chain, Outsourced Design Team, Manufacturer.

Assigned Project Manager and Responsibilities: Project Manager Peter is authorized to lead the project team, assign tasks, approve expenditures within budget, and chair change control board (CCB) as the chairman for change decisions.

Preliminary risks: sensor selection failure, insufficient algorithm accuracy, unstable supply chain, certification delays, etc.

Exit criteria: project acceptance or termination decided by the sponsor.

2. Develop the Project Management Plan

The project management plan integrates all subsidiary plans, including the scope management plan, schedule management plan, cost management plan, quality plan, resource management plan, communication management plan, risk management plan, and procurement plan. The project management plan is a dynamic document that is continuously updated throughout the project based on approval of changes to the document.

Scope Management Plan: Defines how the project scope is defined, developed, and monitored, using the scope statement and WBS as the scope baseline.

Schedule Management Plan: Defines how the project schedule is developed and controlled, using the Gantt chart as the schedule baseline.

Cost Management Plan: Defines how costs are planned, estimated, budgeted, and controlled. A Cost Baseline (\$9,000,000) and a phased cost review process are established. Earned Value Management (EVM) is used for monitoring.

Quality Management Plan: Included in WBS 7 (Testing and verification) and 10.1 (Quality Control), it defines quality standards and verification methods.

Resource Management Plan: It defines the allocation principles for human resources, outsourcing resources, and physical resources of the core R&D team of 20 people and multiple outsourcing suppliers.

Communication Management Plan: It defines who needs what information at what time, and how to transmit it. Stakeholder communication frequency, mechanisms, and emergency event reporting process are defined. Communication mechanisms include daily stand-up meetings, weekly meetings, bi-weekly reviews, and monthly leadership reports.

Risk Management Plan: Risk matrices and mitigation strategies related to technology, data security, market, and supply chain are developed.
Conduct periodic reviews.

Procurement Management Plan: Develop detailed procurement contracts based on the outsourcing sections of the scope statement and resource plan. The contracts include the items to be procured, quantities, and procurement schedule. Define how to externally acquire industrial design, cloud services, and manufacturing services.

Stakeholder Management Plan: Managed by systematically mapping stakeholders using an interest or power grid, identifying key influencers, and developing tailored engagement strategies based on their level of power, interest, and impact on project outcomes.

Change Management Plan: Clearly defines how to submit, evaluate, obtain approval for, and implement changes in the scope statement.

Configuration Management Plan: Implicit in the development process, used to manage hardware and Software product versions.

3. Project Execution

Consistent execution rhythm across teams:

- The hardware, firmware, algorithms, cloud, and APP teams synchronize daily/weekly according to the communication plan.

- Implement an integrated development process: starting with the hardware prototype, complete sensor calibration, then establish the firmware data transmission link, then proceed to AI model development and verification, followed by access to cloud services, and finally achieve the functional display of the APP interface.

Centralized requirements and design management:

- All requirement changes, design documents, and interface specifications are uniformly archived in Jira and Confluence to ensure traceability.
- The system architect periodically checks whether the technical architecture is consistent with the requirements and scope.
- Integrated management of suppliers and outsourcing teams
- The outsourced industrial design team is responsible for appearance and structural design (WBS 3).
- Cloud services and big data development are provided by a third-party vendor (WBS 6.1).
- Trial production and mass production are the responsibility of the manufacturer (WBS 8).
- The project manager is responsible for coordinating delivery schedules, quality control, and contract compliance.

Deliverables management: Tasks are assigned to the team by the project manager based on the WBS and schedule management. The entire team progressively produces product design documents, prototypes, test reports, certification certificates, and the Final Product.

Work performance data management: Retain raw data such as task completion percentage, code commit frequency, and test case pass rate.

Change request management: During implementation, any proposed modifications to the baseline will be submitted as a change request.

4. Project Knowledge Management

Explicit Knowledge:

- Technical documents, test reports, API documentation, and algorithm verification results are all uploaded to the unified knowledge base.
- Share phase achievements through bi-weekly demo reviews.

Tacit Knowledge (Experience, Solutions):

- Facilitate communication through daily stand-up meetings and cross-team problem discussion meetings.

- For key challenges, such as algorithm accuracy, sensor calibration, and cloud latency, create lessons learned documents for subsequent iterations.

5. Monitoring Project Work

Monitoring results are presented in weekly meetings and monthly leadership reports.

Schedule Monitoring: The project manager compares the actual work completed with the planned schedule, and the team reports weekly on completion status, obstacles, and next steps. Formal reviews are conducted at each milestone (P1-P5) to determine whether to proceed to the next phase.

Cost Monitoring: Monthly cost audits are performed according to the cost document to identify budget variances. Earned value management is used monthly to calculate schedule variance and cost variance.

Quality Monitoring: The QA team executes testing and verification reports according to WBS 7.

Risk Monitoring: The risk matrix is updated regularly, with a focus on high risks such as sensor failure and certification failure.

Resources monitoring: Check whether engineering resources meet the peak development demand according to the resource document.

6. Implement Integrated Change Control

All changes must be approved by the Change Control Board (CCB), which includes the project sponsor, project manager, product manager, and technical lead.

Change Process:

Submit a Change Request: Include the scope, budget, resources, schedule, and other relevant details.

Impact Analysis: The project manager organizes an assessment covering the impact on cost, schedule, quality, and risk.

CCB Approval: All significant changes must be approved by the CCB. Any unapproved changes cannot be implemented to prevent scope creep and cost overruns.

Update Baselines: If the change is approved, update the scope, schedule, and budget baselines accordingly, as well as any relevant project documents.

Communicate Change Results: Communicate the results of the change to all relevant stakeholders through weekly meetings, Jira, and email.

7. Close Project or Phase

Conduct phase closure at the end of each major phase (design, development, testing, trial production, mass production), and conduct overall closure after final project delivery.

Closure contents include:

Acceptance Verification: Verify that all deliverables meet the Acceptance Criteria in the Scope document.

Output Archiving: Collect and archive all test results, certification certificates, and UAT results.

Summary Meeting: Hold a project summary meeting and record Lessons Learned. Publish the "Final Project Report" to summarize project Performance.

Knowledge Material Archiving: Complete knowledge material archiving, including code, documentation, verification data, and suppliers files.

Product Handover: Officially hand over the product to the Operations and after-sales team (WBS 8.2).

Contract Closeout: Settle all payments to outsourcing suppliers.

Scope Management (Rui Liu)

1. Scope planning

This plan outlines how the scope of the Smart Bed Project will be defined, confirmed, and controlled. Its purpose is to ensure that the project encompasses only essential work and that any scope changes are managed effectively.

2. Collect requirements

Requirements Gathering Methods:

- **Market Research:** Obtain market requirements through competitive product analysis (WBS 1.1) and user persona research (WBS 1.2).
- **Stakeholder Interviews:** Conduct one-on-one or group interviews with project sponsors, medical advisors, potential users, etc., to uncover deep-seated needs.

Requirements Documentation: All collected requirements have been recorded and integrated into the Comprehensive List of Project Requirements of the Project Scope Statement.

Requirements Tracking: To ensure that no requirements are missed, the project will use a Requirements Traceability Matrix to associate each requirement with corresponding design, development, testing, and other subsequent tasks.

3. Scope definition

The scope baseline consists of the project scope statement and the work breakdown structure (WBS).

Project scope statement

I. General Information

Project Name:	Smart Bed Project	Project No.:	SMART-BED-2025
---------------	-------------------	--------------	----------------

Project Manager:	Amy	Date:	2025/01
------------------	-----	-------	---------

II. Project Overview:

This project aims to R&D a smart bed product integrating multiple sensors and artificial intelligence technologies. The project was launched to meet the growing market demand for intelligent and personalized sleep solutions. The project can help to improve users' sleep quality by providing professional sleep monitoring and intuitive health reports. Then, our project will establish a leading position in the smart home market.

III. Project Goals and Objectives:

Specific: Develop and launch a smart bed with sleep monitoring, temperature and angle control, personalized sleep reports, and sleep apnoea alerts. Customer can use APP to control the bed and check the sleep report. The cloud platform is used for data storage.

Measurable: All hardware and software functions pass all tests defined in WBS Deliverable 7; AI algorithms Sleep staging accuracy >85%; FCC, CE, and UL safety certification are obtained; no high-priority defects are found in User Acceptance Testing (UAT); completion within a 12-month timeframe; project budget controlled within \$9,000,000.

Agreed to: Project goals have been reviewed and agreed upon by the project sponsor, product management team, and core technology leaders.

Realistic: The project goals are achievable based on existing technology reserves and resources.

Time-bound: The project must be completed within 12 months, scheduled to commence in January 2025, with mass production and delivery achieved by December 2025.

IV. Comprehensive List of Project Deliverables:

1. Fully Functional Smart Bed Prototype with integrated multi-sensor system
2. Mobile APP with bed control, sleep report viewing, and alert functions
3. Cloud Platform for data storage, analysis, and personalized report generation
4. AI Sleep Analysis System including sleep staging, apnea detection, and scoring models
5. FCC/CE/UL safety Certification Documents
6. User Manuals & Technical Documentation
7. Production-Ready Design Files for mass manufacturing
8. Final Mass-Produced Smart Bed Units ready for market launch

V. Comprehensive List of Project Requirements:

Business Requirements: The product needs to have a subscription-based payment model, with potential for ongoing revenue through value-added services such as in-depth health analysis.

Functional Requirements:

- Real-time monitoring of user vital signs and sleep patterns.
- Users can control bed angles and temperature via the APP.
- Daily generation and push of personalized sleep quality scores and reports.

- The system should identify potential sleep disorder risks and provide alerts.

Technical Requirements:

- Data must be encrypted during transmission and storage to comply with Privacy protection regulations.
- The bed's mechanical structure must withstand over 100,000 fatigue tests.
- The system's cloud services must guarantee 99.9% availability.
- The mobile APP must be compatible with mainstream iOS and Android operating systems.

VI. Exclusions from Scope:

- To clarify project boundaries, the following items are explicitly excluded from the scope of this project:
- Professional medical treatment-grade diagnostic capabilities are not included. All Health Data and screening results are for health reference only and should not be used as a substitute for professional medical diagnosis.
- This product does not include any disease treatment functions and is intended solely as a monitoring and screening tool.
- Smart home control functions beyond those already defined are not included.
- The production of terminal hardware devices such as mobile phones and tablets is not included.
- Large-scale Marketing campaigns, advertising planning, and execution are not included.

VII. Time and Cost Estimates:

Time Estimation: The total project duration is 12 months. For a detailed breakdown of phases, please refer to the attached project schedule (Gantt Chart).

Cost Estimation: The preliminary project budget is estimated at RMB 4,500,000 – 9,000,000. This estimate will be further refined upon completion of detailed planning.

VIII. Roles and Responsibilities:

Role	Primary responsibilities	Responsible person
Project Manager	Responsible for overall project planning, implementation, monitoring, closure, and stakeholder management; approves the project scope statement and scope baseline; as the CCB chair, approves or rejects major scope changes.	Peter

Product Manager	Defines product requirements and priorities, and is responsible for product direction and acceptance; defines product scope and requirements; assists in the validation of deliverables; as a core member of the CCB, evaluates changes from a product value perspective.	Jack
Systems Architect	Designs the overall technical solution and is responsible for technology selection and architecture decisions; completes work within the approved scope baseline; promptly identifies and reports issues that may trigger scope changes.	Coco
Hardware Team	Responsible for the development and testing of hardware such as sensors and main control boards; completes work within the approved scope baseline; promptly identifies and reports issues that may trigger scope changes.	Candy
Software/ Algorithm Team	Responsible for embedded software, AI algorithms, cloud Platform and APP development; Complete work within the APProved scope baseline; Promptly identify and report issues that may trigger scope changes.	David
Testing Team	Develop test plans, execute all testing activities and submit reports; Complete work within the approved scope baseline; Promptly identify and report issues that may trigger scope changes.	Kitty
Quality and Compliance Team	Ensure products comply with predefined quality Standards and safety Regulations; Complete work within the approved scope baseline; Promptly identify and report issues that may trigger scope changes.	Lee
CCB	Responsible for reviewing, evaluating, approving, or rejecting all scope change requests.	Sophie

IX. Assumptions:

- The supply chain for the core sensors chips required for the project is stable, and there will be no long-term shortages.
- The R&D risks of key technologies, such as multi-sensor data fusion algorithms are within a controllable range.
- A sufficient number of internal and external users can be recruited to successfully complete user acceptance testing.
- Key project personnel will remain stable throughout the project.

X. Constraints:

Time constraint: The project must be completed and launched by December 31, 2025.

Cost constraint: The total project budget shall not exceed \$9,000,000.

Scope constraint: The product functionality scope is subject to Parts IV and V of this document, and any changes must follow formal procedures.

Quality constraint: The product must pass FCC, CE, and UL certification and meet the preset Performance Indicators.

Resource constraint: The core R&D team size is 20 people.

XI. Names and Signatures:

This Statement of Scope has been reviewed and agreed by the following parties and shall serve as the baseline document for project implementation.

Role	Name	Signature	Date
Project initiator	Sophie		
Project Manager	Peter		
Product Manager	Jack		
Technical Lead	Coco		
Other key stakeholders			

XII. Acceptance Criteria:

- All functions have passed the system integration testing and user acceptance testing defined in WBS Deliverable 7, product testing and validation.
- The hardware has passed the durability testing and safety certification (FCC/CE/UL) defined in WBS 7.4.
- The software has no defects of Critical level or higher.
- The accuracy and stability of the algorithms model have reached the preset Indicators.
- The project initiator and the client have signed the acceptance confirmation.

4. WBS

Building upon the WBS structure, we have provided a suggested responsible person for each work package, which will clarify the project's operational guidance.

Smart Bed Project		
Deliverable 1 Project Proposal and Plan	1.1 Market Needs and Analysis Work Package 1.1.1	Competitive Product Analysis Report Writing <i>(Responsible: Product Manager)</i>
	Work Package 1.1.2	Target User Persona and Needs Research <i>(Responsible: Product Manager, Marketing Team)</i>
	1.2 Project Feasibility Study Work Package 1.2.1	Technical Feasibility Assessment (Sensors, Algorithms) <i>(Responsible: System Architect, Technical Lead)</i>
	Work Package 1.2.2	Preliminary Cost Estimation and Return on Investment Analysis <i>(Responsible: Project Manager, Finance)</i>
	Work Package 1.2.3	Preliminary Identification of Project Risks <i>(Responsible: Project Manager, Core Team)</i>
	Work Package 1.2.4	Preliminary Regulatory and Compliance Feasibility (FDA/CE/UL) <i>(Responsible: Quality & Compliance Team)</i>
	2.1 Product Function Design Work Package 2.1.1	Core Function Definition (Data Reports, Posture Adjustment, Temperature Control) and Priority Ranking <i>(Responsible: Product Manager)</i>
	Work Package 2.1.2	Sleep Reports Content, Format, and Personalization Logic Definition <i>(Responsible: Product Manager, Medical Consultant)</i>
Deliverable 2 Product Design	2.2 Technical Architecture Design Work Package 2.2.1	Overall System Architecture Design (Sensing-Processing-Cloud-APP) <i>(Responsible: System Architect)</i>
	Work Package 2.2.2	Sensor Fusion Scheme Technology Model Selection <i>(Responsible: Hardware Team,</i>

	<i>Algorithm Team)</i>	
Deliverable 3 Appearance and Structural Design (<i>Outsourced Team</i>)	3.1 Industrial Design Work Package 3.1.1	Bed Appearance, Material, and Color Design (<i>Responsible: Outsourced Team</i>)
	Work Package 3.1.2	Mattress Physical Button and Indicator Light Design (<i>Responsible: Outsourced Team</i>)
	3.2 Structural Design Work Package 3.2.1	Design of Bed Frame, Substructure, and Adjustable Angle (<i>Responsible: Outsourced Team</i>)
	Work Package 3.2.2	Design of Internal Space for Sensors and Cable Routing within the Bed (<i>Responsible: Outsourced Team</i>)
Deliverable 4 Hardware Development and Integration	4.1 Sensor System Integration Work Package 4.1.1	Selection of Sensors (Pressure, bio-radar, Temperature) (<i>Responsible: Hardware Engineer</i>)
	Work Package 4.1.2	Layout Design and Optimization of Sensors in the Mattress (<i>Responsible: Hardware Team, Mechanical Engineer</i>)
	Work Package 4.1.3	Connection and Wiring Scheme for Sensor Array and Main Control Board (<i>Responsible: Hardware Engineer</i>)
	4.2 Main Control and Electrical System Work Package 4.2.1	Selection of Main Controller (MCU) and Design of Core Circuit Board (PCB) (<i>Responsible: Hardware Engineer</i>)
	Work Package 4.2.2	Power management system and low-power design (<i>Responsible: Hardware Engineer</i>)
	Work Package 4.2.3	Communication module (Wi-Fi/Bluetooth) integration (<i>Responsible: Hardware Engineer</i>)
Deliverable 5 Embedded Software and Algorithms Development	5.1 Embedded Firmware Development Work Package 5.1.1	Sensor driver, data acquisition, and

		synchronization procedures development (Responsible: Embedded Software Engineer)
	Work Package 5.1.2	Equipment-side pre-processing and edge computing logic development (Responsible: Embedded Software Engineer, Algorithms Team)
	5.2 Core AI Algorithms Development	
	Work Package 5.2.1	Multi-sensor data fusion algorithms development (Responsible: Algorithms Team)
	Work Package 5.2.2	Sleep staging, vital signs extraction, and posture recognition model training (Responsible: Algorithms Team)
Deliverable 6 Cloud Platform and Mobile APP Development	Work Package 5.2.3	Daily sleep score model and personalized report generation logic development (Responsible: Algorithms Team, Product Manager)
	Work Package 5.2.4	Sleep disorder (e.g., OSA) screening algorithm development (Responsible: Algorithms Team, Medical Consultant)
	6.1 Cloud Service Development	
	Work Package 6.1.1	User and Equipment Management API Development (Responsible: Backend Development Engineer)
	Work Package 6.1.2	Big Data Storage, Analysis, and Sleep Reports Backend Generation Service (Responsible: Outsourced Team)
	Work Package 6.1.3	Data Security and Privacy Protection Plan Implementation (Responsible: Outsourced Team)
	Work Package 6.1.4	Subscription and payment backend integration (Responsible: Outsourced Team)
	6.2 Mobile APP Development	
	Work Package 6.2.1	APP UI/UX Design (Responsible: UX/UI Designer)

	Work Package 6.2.2	Sleep Reports Display, Historical Trend, and Anomaly Alert Function Development (Responsible: Mobile Development Engineer)
	Work Package 6.2.3	Bed Control (Angle, Temperature Control) Function Development (Responsible: Mobile Development Engineer, Embedded Team)
Deliverable 7 Product Testing and Verification	7.1 Hardware Testing Work Package 7.1.1	Structural strength and durability testing (Responsible: Mechanical engineers, Testing team)
	Work Package 7.1.2	Electrical safety and EMC (Electromagnetic Compatibility) certification testing (Responsible: Outsourced Team)
	Work Package 7.1.3	Sensor accuracy and reliability testing (Responsible: Hardware team, Testing team)
	7.2 Software Testing Work Package 7.2.1	Embedded firmware and cloud API unit testing and integration testing (Responsible: Testing team, Development team)
	Work Package 7.2.2	Mobile APP functionality, Performance, and compatibility testing (Responsible: Testing team)
	Work Package 7.2.3	AI algorithms accuracy and stability testing (Responsible: Algorithms team, Testing team)
	7.3 System Integration Testing Work Package 7.3.1	End-to-end functional testing from sensors to APP (Responsible: Test Team)
	Work Package 7.3.2	Sleep reports accuracy comparison and verification (with professional medical equipment) (Responsible: Test Team, Algorithms Team, Medical Advisor)
	7.4 Certification and Durability Testing Work Package 7.4.1	Safety certification (FCC/CE/UL) application and passing (Responsible: Outsourced Team)

	Work Package 7.4.2	Complete machine lifetime and fatigue testing (<i>Responsible: Test Team</i>)
	7.5 User Acceptance Testing (Pre-Launch Validation)	
	Work Package 7.5.1	Small batch trial production of prototype machines (<i>Responsible: Production Team</i>)
	Work Package 7.5.2	Organize internal and external users for experience testing (<i>Responsible: Product Manager, UX Researcher</i>)
	Work Package 7.5.3	Collect feedback and drive product iteration and optimization (<i>Responsible: Product Manager, Project Manager</i>)
Deliverable 8 Production and Delivery (<i>Outsourced Team</i>)	8.1 Production Preparation	
	Work Package 8.1.1	Supply chain establishment and material procurement (<i>Responsible: Supply Chain Team</i>)
	Work Package 8.1.2	Production line setup (<i>Responsible: Production Engineer</i>)
	Work Package 8.1.3	Development of production process flow and quality control standards (<i>Responsible: Production Engineer, Quality Team</i>)
	Work Package 8.1.4	Personnel training for production (<i>Responsible: Production Team</i>)
	Work Package 8.1.5	Pilot production run (PVT) and issue resolution (<i>Responsible: Production Team, Engineering Team</i>)
	8.2 Mass Production and Delivery	
	Work Package 8.2.1	Supply chain management and large-scale production (<i>Responsible: Supply Chain Team, Production Team</i>)
	Work Package 8.2.2	Establishment of packaging, logistics, and after-sales service system (<i>Responsible: Operations Team, Customer Service Team</i>)

	Work Package 8.2.3	Warranty policy, repair process, and customer support training (<i>Responsible: Customer Service Team</i>)
Deliverable 9 Post-Launch Monitoring & Continuous Improvement	9.1 User feedback Collection and Bug Tracking	
	Work Package 9.1.1	Establish real-world user feedback channels (APP, survey, customer service) (<i>Responsible: Product Manager</i>)
	Work Package 9.1.2	Collect and categorize bug reports from live users (<i>Responsible: QA Team, Development Team</i>)
	Work Package 9.1.3	Prioritize and assign feedback-driven fixes and enhancements (<i>Responsible: Product Manager, Project Manager</i>)
	9.2 System Performance Monitoring	
	Work Package 9.2.1	Set up monitoring tools for server load, response time and uptime (<i>Responsible: Operation Team, Cloud Team</i>)
	Work Package 9.2.2	Analyze system stability, error rates and data transmission late (<i>Responsible: Cloud Team</i>)
	Work Package 9.2.3	Generate regular performance and reliability reports (<i>Responsible: Operation Team, Project Manager</i>)
	9.3 Product Update and Iteration Planning	
	Work Package 9.3.1	Plan APP function updates based on user and performance data (<i>Responsible: Product Manager, Development Team</i>)
	Work Package 9.3.2	Implement version control, regression testing and development schedule (<i>Responsible: Development Team, QA Team</i>)
	Work Package 9.3.3	Release updated versions and notify improvements to users (<i>Responsible: Marketing Team, Product Manager</i>)
Deliverable 10 Project Management and	10.1 Project Control Work Package 10.1.1	Schedule, cost, and quality monitoring

Support (Throughout the process)		<i>(Responsible: Project Manager)</i>
	Work Package 10.1.2	Risk and issue management <i>(Responsible: Project Manager)</i>
	Work Package 10.1.3	Change Management <i>(Responsible: Project Manager, Change Control Board)</i>
	Work Package 10.1.4	Project status reports and meetings <i>(Responsible: Project Manager)</i>
	10.2 Team and Communication	
	Work Package 10.2.1	Team building and management <i>(Responsible: Project Manager, Functional Manager)</i>
	Work Package 10.2.2	Internal and external communication plan implementation <i>(Responsible: Project Manager)</i>

5. Scope Verification

Verification Timing: Conduct verification after each major phase or deliverable is completed, especially after testing and pilot production.

Verification Method:

- The project manager submits deliverables and related test reports.
- The project sponsor, product manager, and customer representatives conduct a review.
- Upon successful review, relevant parties formally sign the project acceptance documents.

Key Confirmation Points: Product design documents, fully functional prototypes, test reports, safety certification certificates, and final mass production products.

6. Scope change control

Change Request Submission: Any stakeholder may submit change requests in writing, such as via a Change Request Form.

Change Request Assessment: The project manager is responsible for convening a change impact assessment, which should cover:

- Impact on scope: Whether the WBS needs modification.
- Impact on schedule: Refer to the schedule to assess potential delays.
- Impact on cost: Assess budget increases or decreases.

- Impact on resources: Assess whether additional personnel or equipment are required.
- Impact on technical feasibility and quality.
- Impact on project risks.

Change Request Approval:

- All change requests must be submitted to the Change Control Board (CCB) for approval.
- Suggested CCB members: Project Sponsor, Project Manager, Product Manager, Technical Lead.

Change Implementation:

- Once a change is approved, the project manager needs to update the scope baseline (i.e., update the WBS, scope statement, and schedule).
- Communicate the change outcomes to all stakeholders.

Change Rejection: The project team is not authorized to execute any change requests that have not been approved through the formal change control process.

Continuous Monitoring: During project status meetings, regularly compare actual work against the WBS to ensure the team is not performing work outside the scope of the WBS.

Quality Management (Kanokkan)

Quality Management in the Smart Bed Project ensures that the final product is safe, reliable, and capable of providing accurate sleep insights to users. Because the system involves sensors, firmware, AI models, cloud services, and a mobile application, quality must be monitored throughout the entire development process. Moreover, careful planning and continuous reviews help the team identify issues early, which reduces the risk of major defects later. Therefore, this section describes the quality objectives, assurance activities, testing approach, acceptance criteria, and key metrics used in the project.

1. Quality Objectives

The main quality objectives are:

1. **Sensor Accuracy:** Sensors must produce stable readings with an error margin of $\pm 5\%$, **which** ensures that sleep and health data remain trustworthy across different conditions.
2. **AI Performance:** The AI model must reach at least 85% accuracy, and furthermore, it should perform consistently when validated against medical-grade data.
3. **System Reliability:** Firmware, cloud, and mobile app components should operate smoothly together without critical failures, as a result creating a dependable user experience.
4. **Safety & Compliance:** All hardware must pass FCC/CE/UL certification, which demonstrates that the device is safe for daily use.
5. **User Experience:** The app must be easy to navigate, visually clear, and show sleep information in a simple format; moreover, it should maintain a low crash rate.

2. Quality Assurance (QA)

Quality Assurance focuses on preventing problems before they appear. Furthermore, QA activities help the team build quality into the product at every stage.

Key QA activities include:

- **Requirements Review:** Ensuring each requirement is clear, complete, and testable, **which** prevents confusion during development.
- **Design Review:** Checking sensor placement, electrical design, and system architecture to avoid design flaws.
- **Prototype Evaluation:** Testing early hardware and firmware to understand how sensors behave in real settings.
- **Code Review & Continuous Integration:** Reviewing code quality and using automated tools to detect issues early.
- **Pre-Compliance Tests:** Running basic safety checks before official certification, **therefore** reducing the risk of failing later.
- **UI/UX Review:** Ensuring the app is simple and intuitive, which helps users understand their sleep data easily.

3. Quality Control (Testing)

Quality Control verifies that the final output meets the required standards. In contrast to QA, which focuses on prevention, QC focuses on detecting defects.

Testing activities include:

Hardware Testing

- Checking sensor calibration and response under different weights and temperatures.
- Running durability and stress tests.

Firmware Testing

- Verifying that data is collected and transmitted correctly.
- Ensuring communication between all components remains stable.

AI Model Testing

- Measuring sleep stage accuracy and the ability to detect abnormalities.
- Adjusting thresholds to avoid false alerts.

Cloud & Backend Testing

- Ensuring data moves correctly through APIs.
- Testing security and response time.

Mobile App Testing

- Checking for crashes, UI problems, and incorrect visualizations.

System Integration Testing

- Confirming that the full process (sensor → firmware → cloud → app) works without interruption.

User Acceptance Testing (UAT)

- Inviting test users to try the product and give feedback, which helps improve the final release.

4. Acceptance Criteria

The Smart Bed will be considered acceptable when:

- All test cases pass with no critical or high-priority defects.

- Sensors remain accurate within the required range.
- AI accuracy reaches **85% or higher**.
- The device passes FCC/CE/UL safety certifications.
- The mobile app runs smoothly with minimal crashes.
- Sleep reports load correctly and on time.
- Users in UAT can operate the system without difficulty.

Moreover, these criteria ensure the product meets both technical and user expectations.

5. Quality Metrics

To measure progress, the project will track:

- Sensor error rate (%)
- AI model accuracy (%)
- Number of defects per testing cycle
- Test coverage percentage
- App crash rate
- Cloud response time

Furthermore, these metrics help the team evaluate performance and identify areas needing improvement.

In conclusion, Quality Management ensures the Smart Bed is safe, accurate, and reliable. Moreover, combining Quality Assurance (prevention) with Quality Control (testing) creates a strong process that supports consistent performance. As a result, the project team can deliver a product that meets industry standards and provides a positive user experience.

Quality Management - Testing (Kanokkan)

1. Project Manager – Teammate A

Department: Integration, Schedule, Cost, Risk

Responsibilities:

- Build and maintain the **project plan**, scope baseline, and Gantt schedule
- Coordinate **all departments** (hardware, software, AI, app, testing)
- Approve requirements, testing plans, prototypes
- Control changes through the **Change Control Board (CCB)**
- Conduct weekly meetings & milestone reviews
- Track cost vs. budget using Earned Value Management (EVM)
- Maintain stakeholder communication plan

Tools Used:

- MS Project / Gantt Chart
- Jira / Confluence for task management
- Risk Register
- Cost Tracking Sheet (EVM)

Deliverables:

- Project Charter
- Scope, Schedule, Cost Baseline
- Weekly Status Reports
- Change Request Log
- Final Project Report

2. Product Manager – Teammate A

Department: Product Requirements, UI/UX, User Stories

Responsibilities:

- Define core product features (Sleep Score, Temperature Control, Alerts)
- Write **user stories** with acceptance criteria
- Validate design with user personas (e.g., elderly users, professionals)
- Support UAT and collect user feedback
- Prioritize requirements using MoSCoW / backlog grooming

Tools Used:

- Figma (UI wireframes)
- User Story Map
- Requirements Traceability Matrix

Deliverables:

- Product Requirement Document (PRD)
- User Stories & Acceptance Criteria
- UI flow diagrams
- Feature Prioritization List

3. System Architect – Teammate B

Department: End-to-End Architecture (Sensors → Firmware → AI → Cloud → App)

Responsibilities:

- Design system architecture for sensing, processing, data flow, and app
- Select frameworks and technologies
- Create interface contracts (API between firmware/cloud)
- Approve hardware layout compatibility with software flow
- Conduct integration checks during early testing

Tools Used:

- Architecture Diagrams

- UML sequence diagrams
- Data Flow Mapping

Deliverables:

- Full System Architecture Document
- Sensor–Firmware–Cloud Data Model
- Interface Specification Sheet

4. Hardware Engineering Team – Teammate C

Department: Sensors, PCB, Electrical Systems

Responsibilities:

- Select and test sensors (Pressure, Bio-radar, Temperature, Motion)
- Develop PCB, controller board, and wiring design
- Implement power management system
- Ensure safety standards for FCC/CE/UL
- Conduct durability and structural tests

Tools Used:

- Sensor Calibration Bench
- PCB Design Software (Altium, Eagle)
- Multimeter, Oscilloscope

Deliverables:

- Hardware Prototype (Alpha & Beta)
- Sensor Calibration Reports
- PCB Schematics
- Safety Verification Reports

5. Embedded Firmware Team – Teammate D

Department: Firmware Code, Data Synchronization

Responsibilities:

- Build drivers for sensors
- Implement real-time data capture and buffering

- Optimize battery usage (low-power design)
- Develop communication between mattress and cloud/app
- Support system integration testing

Tools Used:

- C/C++, RTOS
- Embedded Debugger
- Logic Analyzer

Deliverables:

- Firmware v1, v2, certification-ready firmware
- Data Synchronization Module
- Embedded Pre-Processing Functions

6. AI / Algorithm Team – Teammate E

Department: Machine Learning, Sleep Analytics

Responsibilities:

- Build multivariate sensor-fusion model
- Train algorithms for sleep stage classification
- Develop Sleep Score Model
- Detect anomalies (apnea, abnormal HR, irregular breathing)
- Validate models using medical devices

Tools Used:

- Python, TensorFlow/PyTorch
- GPU servers for training
- Confusion Matrix, ROC curves

Deliverables:

- Trained AI Model (Sleep Staging + Anomaly Detection)
- Sleep Score Algorithm
- Accuracy Validation Report

- Model Documentation

7. Cloud & Backend Team – Teammate F

Department: Cloud Infrastructure, API, Data Storage

Responsibilities:

- Build backend APIs for user account, report generation, device data
- Store sleep logs securely
- Build subscription + payment modules
- Implement encryption and privacy compliance
- Ensure uptime monitoring (99.9% target)

Tools Used:

- AWS/GCP/Azure
- Database (PostgreSQL/MongoDB)
- API Gateway
- Security tools (OAuth2, encryption)

Deliverables:

- Cloud Backend System
- API Documentation
- Security and Privacy Compliance Report
- Payment System Integration

8. Mobile App Development Team – Teammate G

Department: Mobile UI/UX + Bed Control Features

Responsibilities:

- Create app screens for Sleep Score, trend charts, alerts
- Implement bed controls (angle, temperature)
- Integrate with backend APIs and firmware signals
- Ensure compatibility across iOS & Android

Tools Used:

- Figma

- Swift, Kotlin, Flutter
- Mobile Testing Frameworks

Deliverables:

- Mobile App v1 UI
- Sleep Report Dashboard
- Bed Control Interface
- App Release Build (TestFlight / Google Play Beta)

9. Testing & QA Team – Teammate H

Department: Hardware + Software + System Testing

Responsibilities:

- Write test cases for hardware, firmware, cloud, app
- Conduct unit tests, integration tests, system tests
- Run medical-accuracy comparison testing
- Execute UAT and gather feedback
- Sign off final quality reports

Tools Used:

- Test Case Management Tools
- Regression Testing Suite
- Various sensors compared with medical devices

Deliverables:

- Test Plan + Test Case Matrix
- Integration Test Report
- UAT Report
- Certification Readiness Report

10. Procurement, Manufacturing & Operations – Teammate I

Department: Supplier Management, Production, Logistics

Responsibilities:

- Source suppliers for sensors, PCB, mattress materials
- Oversee industrial design vendor & manufacturing partner
- Manage pilot production (PVT)
- Plan packaging, logistics, after-sales support

Tools Used:

- Supplier Scorecard
- BOM Cost Analysis
- Procurement Contracts

Deliverables:

- Supplier Evaluation Sheet
- Production Plan
- Packaging & Logistics Plan
- Warranty + After-Sales Flow

Stakeholder Management

1.introduction

Since the project touches various areas including sensor R&D, embedded software development, algorithm development, mobile app development, structural engineering, and regulatory certification, it has various stakeholders with diverse interests and expectations. This becomes very important to handle effectively to achieve various goals including technical viability, acceptance, and successful launch in the marketplace. This document provides a full-fledged stakeholder management plan for dealing with this project.

2. Stakeholder Identification

For Smart Bed Project, stakeholders can be categorized into internal and external stakeholders. Internal stakeholders include individuals within the company directly involved in development, engineering, or project management. External stakeholders include consumers, government agencies, manufacturers, and other project partners.

2.1 Stakeholder Categories

The table below provides an overview of the major stakeholder groups and their primary roles in the project.

stakeholder Category	Examples	Primary Interests
Internal	Project Manager, Product Manager, Hardware Team, Algorithm Team, Testing & QA, Compliance Team	Technical success, timeline adherence, product performance
External	End Users, Beta Testers, Regulatory Agencies (FDA/UL/CE), Manufacturing Partners, Investors, Medical Advisors	Safety, accuracy, usability, compliance, return on investment

Each stakeholder has its own set of expectations. End-users are concerned with comfort, data privacy, and providing valuable health information, while government agencies are concerned with safety standards and medical device regulation.

3. Stakeholder Interest and Influence Analysis

Stakeholders can exercise varied types and levels of influence on projects and also can have varied interests in projects. By understanding this, project teams are able to allocate project attention and communication resources to projects appropriately.

3.1 High-Influence, High-Interest Stakeholders

The stakeholders include Project Managers, Product Managers, System Architects, Hardware Engineers, Algorithm Teams, and Compliance Teams. Their involvement defines essential parameters related to how to design the system architecture, how accurate the AI model is, how to integrate sensors, how safe the system is, and how to manage projects. They need to be properly managed and updated on matters pertaining to project developments.

3.2 High-Influence, Low-Interest Stakeholders

This category belongs to investors and manufacturing partners. They possess great power since they allocate resources and can manufacture, but they are not concerned with technological matters. They also require updates on matters pertaining to finance and business milestones.

3.3 Low-Influence, High-Interest Stakeholders

Usability, ease of use and data protection is a big concern for end users, support staff and beta testers. Although their influence on products and services is unlikely to be immediate, their feedback is important for their improvement for user acceptability.

3.4 Low-Influence, Low-Interest Stakeholders

General ecosystem partners usually need only occasional updates or monitoring as their role is minimal.

4. Stakeholder Engagement Strategy

A way to engage stakeholders in right ways during the course of a project. The techniques employed to engage the stakeholders rely on their function and the anticipated level of contribution.

4.1 Engagement with Technical Teams

The technical teams, which include hardware engineering, embedded software development, cloud backend development, mobile application development, and AI algorithm development, form the operations core for this project. Their involvement encompasses:

- # Organize weekly cross-departmental standing meetings to coordinate project progress.
- # “Technical design review, including architecture design, sensor configuration, and algorithmic structure.”
- # Share documents and coordinate hardware and software subsystems.
- # Apply defect categorization and the iterative cycle for ensuring continuous integration and testing.

That design is therefore vital in abating technological risks and allowing balanced developments of complicated subsystems.

4.2 Engagement with Compliance and Regulatory Stakeholders

Smart beds can also be embedded with various other medical functionalities, including respiration monitoring and vital signs surveillance, among many others. Regulatory approval needs to be sought for this product, which would involve:

- # Expert consulting services on merit-based considerations
 - # Design documentation to support regulatory evaluation
 - # Preliminary Analysis on UL, CE, FCC, or FDA Requirements
 - # Reviewing stages of prototypes, beta versions, or pre-production versions
- Active involvement in this kind of process also prevents problems created by delays in certification or costly redesigns.

4.3 Engagement with End Users and Beta Testers

Involvement by participants is necessary to accurately report on matters pertaining to the clarity of sleep and comfort, sensor quality, and privacy concerns. The strategies employed include the following:

- # Perform beta testing with over 100 participants.
- # Embed a feedback mechanism in the app to gather data
- # Learn to report sleep and conduct UX interviews
- # Employ A/B testing for feature releases and interaction design

This also ensures active engagement of users to assure the contents of the reports match their expectations.

4.4 Engagement with Investors and Executive Stakeholders

The stakeholders mentioned above should be made to understand the risk, profits or losses and schedule risk:

Executive Briefing

Budget and Milestone Report

Effective communication about technical risks and mitigation strategies

Maintaining confidence at the executive level enables ensuring resources.

4.5 Engagement with Supply Chain and Manufacturing Partners

As products move toward production, communication with parties involved in their manufacturing becomes increasingly important. The basic works are:

Design for Manufactureability Review

Participate in development activities regarding BOM from an early stage of Bill of Material.

Assessment of readiness of factories before trial production.

It ensures the continuation of communication throughout any mold, molding, or production stage.

This approach guarantees that the production is manufacturable, efficient, and reliable.

5. Communication Plan

The basic elements are the following:

Share project updates with the internal development team on a weekly basis.

Meetings for reviewing milestones and risks are held every X months.

Perform checks for compliance with law partners every two months.

Leverage mobile technology to obtain continuous feedback from customers.

List out the assessment on manufacturing readiness during the pre-production stage. -

The communication plan has been created to ensure traceability, consistency, and flexibility to accommodate project development changeover..

6. Stakeholder Risk Assessment

Risks that are related to stakeholder factors can affect the project execution, viability, or approval stage. Some risks are:

Approval by regulation

Hurdles in complying with FDA/UL/CE standards might influence the timing of product market entry.

First, get in touch with compliance staff, independent regulatory consultants and document preparers.

Sensor Error & Algorithm Issues

The credibility of the product may be affected by its inadequate accuracy.

Adoption of multi-sensor fusion technology, verification with medical equipment and iterative calibration are the mitigation measures.

User worries about data protection

Data related to health and sleep require strict security protocols.

Take steps to encrypt data, calculate locally, and be transparent with the data user.

Manufacturing irregularities

Low output or supply chain disruptions can affect the release of products.

Countermeasures: trial production tests and monitoring of production partners. Lack of coordination between various teams can cause variances between hardware, software, or artificial intelligence development teams. Countermeasures: develop coordination between teams and dependency mapping.

Risk control adds to project stability and enables smooth execution of every project stage.

Procurement Management

1. Introduction

Smart Bed Project embraces various advanced technologies, including hardware engineering, sensor technology, embedded systems, artificial intelligence algorithms, cloud computing, mobile applications, and certification procedures. On other fronts, it also lays great stress on procurement management to ensure timely availability, within affordable boundaries, and within required quality standards for every element or service.

The success of this project is also dependent on factors other than technology realization, including supply chain viability, legislation, and manufacturability. As part of technology realization, procurement has to also be integrated with R&D and procurement costs. Intelligent bed projects are part of procurement management and involve various procurement activities related to demand generation, implementation progress, target costing, suppliers, and risk factors.

2. Procurement Requirements Identification

Per the WBS of the project, technical specification, system architecture and the development phases of the Smart Bed Project, procurement requirements can be classified into 5 categories.

2.1 Sensor Components

These represent the core technological foundation of the smart bed and include:

- # Pressure sensor arrays
- # Bio-radar units for respiration and heart rate monitoring
- # Temperature and humidity sensors
- # Fiber-optic or flexible sensing materials
- # Motion and posture detection sensors

These components are high-risk, high-cost items with stringent performance requirements.

2.2 Electronic Control Components

These include:

- # MCU (main controller)
- # PCB boards

- # Power management modules
- # Wireless communication modules (Wi-Fi/Bluetooth)
- # Cables, connectors, and shielding materials

Because the bed must pass FCC/CE/UL certification, the selected components must meet strict safety and electromagnetic compatibility requirements.

2.3 Mechanical and Structural Parts

Including:

- # Bed frame and internal support structures
- # Adjustable angle mechanism
- # Foam materials for mattresses and protection fabric
- # Cable Routing Components & Internal Compartments

Such elements need to possess qualities of endurance, support ability, and stability.

2.4 Software and Third-Party Services

Although the company develops its core software, it outsources certain services to a variety of external firms including:

- # Cloud hosting and storage
- # Database and back-end support
- # Subscription and payment gateway services
- # Solutions for network protection and data encryption

Medical consultation in order to verify algorithms

2.5 Certification and Testing Services

Including:

- # FCC/CE/UL certification services
- # EMC (Electromagnetic Compatibility) testing
- # Structural fatigue and durability testing
- # Medical-grade comparison studies for vital-sign accuracy

These services are essential for market readiness and regulatory compliance.

3. Procurement Schedule (Integrated with Project Timeline)

Procurement activities must be completely aligned with whichever phase of the project development in order to ensure timely delivery of materials and services. According to the General Progress Program, procurement work will be carried out in a systematic manner through the following ways by the project team:

Phase 1: Research & Concept (Month 1–2)

Procurement Focus:

- # Initial sensor samples (pressure sensors, bio-radar prototypes)
- # IMCU engineering samples
- # Preliminary testing equipment

Purpose: Feasibility studies, PoC validation.

Phase 2: Product & Technical Design (Month 3–4)

Focus of Procurement:

- # First sensor sample (pressure sensor, biological radar prototype)
- # Integrated Circuit Microcontroller engineering sample
- # Preliminary test equipment

Purpose: Feasibility study & verification of concepts.

Phase 3: Hardware & Software Development (Month 4–8)

Focus of Procurement:

- # Second round of sensor samples from various suppliers
- # Prototype structural elements (structural materials, foam, paint)
- # I3D or CNC structural prototype

Purpose: To contribute to the development of industrial design, structural design, and system architecture.

Phase 4: Testing, Certification & Pilot Production (Month 8–11)

Focus of Procurement:

- #Batch test parts (over 100 pieces)
- #Assembly tools, fixtures, and trial production equipment
- #Certification services (FCC/CE)
- # Durability and medical comparison services

Purpose: This is to ensure that the device performs well in its environment and to start mass production.

Phase 5: Mass Production & Launch (Month 11–12)

RICE Analysis:

- # Mass production materials (sensors, electronics, structural parts)
- # Packaging materials
- # Logistics and warehousing services

Spare parts for warranty and post-sale services

Purpose: To ensure production capacity and readiness for launch.

4. Procurement Budget Planning (Category-Based)

The smart bed project uses a classified cost structure instead of soliciting quotations from particular suppliers as per regular project procurement practices. The budget is divided into the following cost categories:

4.1 Sensor Components (25–30% of Total Cost)

Because sensors determine the system's accuracy and reliability, they represent the largest cost segment.

4.2 Electronics and PCB Systems (15–20%)

Including MCU, PCB, wireless modules, and power management.

4.3 Mechanical and Structural Parts (20–25%)

Bed frame materials, adjustable mechanisms, mattress foam, and protective fabrics.

4.4 Software, Cloud, and External Services (10–15%)

Cloud hosting, cybersecurity, subscription system integration, and medical consultations.

4.5 Testing & Certification (10–12%)

FCC/CE/UL certification, EMC compliance testing, and machine durability testing.

4.6 Pilot Production & Manufacturing Preparation (5–8%)

This includes tooling, production fixtures, trial production in small batches, and quality control settings. The budget framework is clearer than other sources. It reflects the usual distribution of costs of IoT medical products.

5. Supplier Selection and Evaluation

The supplier selection also needs to focus on their technical capabilities, reliability, and ability to carry out contract work within the specified constraints on costs and certification. The framework for supplier evaluation can be described as below:

5.1 Technical Capability (Weight: 30%)

Ability to meet accuracy requirements

Experience with medical or smart home sensors

Availability of engineering samples and technical documents

5.2 Cost Efficiency (Weight: 25%)

- # Unit price and cost structure
- # Long term – Cost reduction potential
- # Minimum Order Quantity Requirements

5.3 Delivery Capability (Weight: 20%)

- # Cycles for delivering goods or services to or
- # Scalable production capacity
- # Ability to fulfill the delivery schedule for alpha and beta testing stages

5.4 Quality and Compliance (Weight: 20%)

- # ISO certifications
- # Component-level regulatory compliance
- # Historical yield rate and quality data

5.5 Flexibility and Collaboration (Weight: 5%)

- # Support for rapid prototyping
- # Responsiveness to design changes

This deductive and systematic risk assessment and greater mitigation strategy diminishes exposure in supply chains while also assuring that suppliers have the capacity to support the project from prototypes to mass production.

6. Procurement Risks and Mitigation Strategies

The procurement risk has a direct bearing on project schedule and cost performances.

The major risks and mitigations are as below:

6.1 Delay in Key Sensor Delivery

- # Impact: Delays to Alpha and Beta testing
- # Mitigation: Dual suppliers, early sample purchase, phased inventory locking

6.2 Cost Overrun in Material Procurement

- # Impact: Exceeds BOM cost target
- # Mitigation: Cost review at Phase 3, annual pricing agreements, DFM optimization

6.3 Certification Failure

- # Impact: Delays mass production and market launch
- # Mitigation: Use pre-certified components, conduct pre-compliance testing

6.4 Supplier Capacity Limitations

- # Impact: Cannot meet mass production demand

Mitigation: Conduct PVT tests, prioritize suppliers with multiple production lines

6.5 Supply Chain Instability

Impact: Component shortages or long lead times

Mitigation: Multi-sourcing strategy, early purchasing for high-risk items

This will also help mitigate risks associated with depending on 'single sourcing' and risks related to 'sudden interruptions'. This will enable stability of project progress and ensure procurement deliveries matching project milestones as well.

7. Conclusion

As far as the success factors for the smart bed project are concerned, they are sensor technology, technology integration, design, and regulatory compliance testing. To procure resources for evaluation within the project's timeframe and spending needs, timely provision for resources is important, ensuring that the transition phase for process evaluation can begin.

Furthermore, efficient procurement strategies also improve supply chain resilience as well as readiness for mass production and distribution for the project. Demand analysis, category budgeting, supplier evaluation, risk evaluation are key factors in designing procurement strategy for the project. In conclusion, procurement management plays an important role for success of smart bed project in terms of technology, regulation, and feasibility.

Resource Management (Ying Ou)

1. Resource Categories

Role	Key Responsibilities & WBS mapping
Project Management	Project Proposal & Feasibility Phase 1.2.2 Conduct preliminary cost estimation and ROI analysis 1.2.3 Lead early project risk identification 1.3.1 / 1.3.2 Develop project charter and preliminary project plan Product Testing Phase 7.5.3 Drive user feedback–based product iteration during UAT Post-Launch Phase 9.1.3 Prioritize feedback-driven bug fixes with QA & dev 9.2.3 Co-author performance/reliability reports 9.3.3 Approve release notes and update communication 10 Project Control
Product Manager	Market Research 1.1 Market Research Design 2.1 Product Design Algorithm Development 5.2.3 Co-develop sleep score model & report logic Testing&UAT 7.5.2 Organize internal/external user testing 7.5.3 Drive product iteration based on feedback Post-Launch 9.1.1 — Maintain real-world feedback channels 9.1.3 — Prioritize fixes and improvements

	9.3.1 — Plan app function updates 9.3.3 — Communicate new releases with Marketing
System Architect	Technical Feasibility & Architecture 1.2.1 Technical Feasibility Assessment 2.2 Technical Architecture Design Testing 2.3.4 Integration smoke test (sensor → firmware → basic model output)
Hardware Engineers	4.1 Sensor System Integration Electrical System Development 4.2 Main Control and Electrical System Testing 2.3.1 Early prototype functional testing 2.3.2 Sensor calibration & validation 7.1.3 Sensor accuracy and reliability testing
Mechanical Engineers	Sensor & Structural Integration 4.1.2 Support sensor layout in mattress structure 1.3.2.1 (outsourced design reference) Provide engineering review when needed Testing 7.1.1 Lead structural strength & durability testing
Embedded Firmware Engineers	Early Prototype & Testing 2.3.1 Early prototype functional testing 5.1 Embedded Firmware Development App Integration 1.6.2.3 Collaborate on bed-control (angle/temperature) app functions
AI/Algorithm Engineers	Early Prototype & Testing 2.3.2 Sensor calibration & validation 2.3.3 Early AI unit tests (staging, signal extraction from simulated inputs)

	2.3.4 Integration smoke test (sensor → firmware → basic model output) AI Model Development 5.2 Core AI Algorithms Development Software Testing 7.2.3 Test accuracy & stability of AI models System Verification 7.3.2 Validate sleep report accuracy with medical devices
Backend/Cloud Engineers	6.1 Cloud Service Development Post-Launch Operations 9.2.1 Deploy monitoring tools with Operations 9.2.2 Analyze system stability, errors, and latency
Mobile App Developers	App Feature Development 6.2.2 Build sleep report display, trends, and alerts 6.2.3 Implement bed control (angle/temperature) Post-Launch 9.3.1 Implement new app features in update cycles 9.3.2 Support regression testing with QA
QA/Testing Team	Hardware Testing 7.1.1 Structural testing 7.1.3 Sensor accuracy testing 7.2 Software Testing 7.3 System Testing Post-Launch 9.1.2 Categorize bug reports from live users
Operations Team	Production & Delivery Support 8.2.2 Build logistics & after-sales system

	Monitoring & Maintenance 9.2.1 Server monitoring setup 9.2.3 Issue regular system performance reports
Finance Team	Feasibility & Compliance 1.2.2 Preliminary Cost Estimation and Return on Investment Analysis 1.2.4 Support certification & regulatory feasibility work
Compliance/Quality Team	Regulatory Preparation 1.2.4 Evaluate feasibility for CE/FCC/UL/FDA Testing Oversight 7.1.2 Support outsourced EMC & safety certification testing

2. Coordination with Outsourced Vendors

To ensure smooth integration between internal work and outsourced deliverables, the Resource Management plan includes three major coordination responsibilities:

- Requirements Handover
 - The internal project manager and engineering leads prepare and deliver detailed technical specifications to outsourced vendors.
 - Including clarifying functional expectations, constraints, and integration needs to minimize misunderstandings and reduce rework.
- Integration Management
 - Internal development teams evaluate and integrate outsourced outputs—such as interface elements, backend services, or production documentation—into the overall system architecture.
 - They ensure these external components function correctly with internally developed hardware, firmware, and algorithms.
- Quality Assurance Alignment
 - The internal QA team is responsible for validating whether outsourced deliverables meet project quality standards.
 - Including checking API behavior, performance readiness, regulatory preparation, and overall compatibility with the Smart Bed's technical components.

3. Resource Requirements by Project Phase

Phase	Required Resources	Description
Phase 1	PM, Product Manager, System Architect, Finance/Compliance	Mainly analytical personnel
Phase 2	Product Manager, System Architect, Hardware & Algorithm leads, Outsourced Industrial Design Team	Cross-functional collaboration
Phase 3	Hardware Team, Embedded Team, Algorithm Team, Backend, Mobile development Team	Peak workload — largest engineering resource use
Phase 4	QA Team, Hardware/Software Engineers, Certification Lab	cross-team availability
Phase 5	Manufacturing Vendor, Supply Chain Team, Operations	Mostly external vendor

Cost Management (Ying Ou)

1. Cost Estimate (Kanokkan & Ying Ou)

Cost Category	Sub-Item	Key Components	Estimated Cost Range(USD)		Rationale
			Sub-Item	Total	
Salaries (Internal Labor)	Engineering	• Hardware Engineers • Firmware/Embedded Developers • AI/ML Engineers • Backend/Cloud Engineers • Mobile Developers (partially in-house) • System Architect	\$1,200,000 – \$2,000,000	\$1,800,000 – \$3,000,000	Engineering drives all core technical deliverables (sensors, PCB, firmware, AI, cloud). Requires long-term full-time involvement across design, development and integration phases.
	Product & Project Management	• Product Manager • Project Manager (schedule, risk, integration oversight)	\$250,000 – \$450,000		Ensures product requirements, coordination with vendors, timeline control and cross-functional alignment across all phases.
	QA & Testing Team	• QA testers (software & firmware) • Hardware reliability testing • Integration & regression test engineers	\$150,000 – \$300,000		Required for continuous testing from prototype to certification, ensuring system stability, safety and app reliability.
	Operations & Support Staff	• Operations Team (production & vendor coordination) • Customer Support (launch preparation)	\$200,000 – \$250,000		Late-phase coordination with outsourced manufacturing, logistics and market launch.

Materials (Hardware Components & Prototype Builds)	Sensors	• Temperature sensors • Pressure sensor array • Bio-radar modules • Sensor calibration tools	\$120,000 – \$300,000	\$500,000 – \$1,200,000	Core sensing system for sleep tracking, apnea detection, posture recognition. Includes multiple iterations for prototypes and pilot builds.
	PCB & Electrical Components	• Custom PCB fabrication • MCU (main controller) • Wi-Fi/Bluetooth modules • Power management chips • Connector & wiring components	\$80,000 – \$200,000		For firmware logic, sensor integration., wireless communication.
	Mechanical Materials	• Bed frame structure • Actuators for angle adjustment • Mattress integration • Sensor housing & protective casing	\$150,000 – \$350,000		For industrial design, structural stability and ergonomic requirements. Includes materials for several design iterations.
	Prototype & Pilot-Run	• Alpha prototypes • Beta prototypes • Pilot production • Test equipment & consumables	\$150,000 – \$350,000		Multiple rounds of prototypes required for hardware verification and certification readiness.
	Industrial Design Outsourcing	• Exterior design • Mattress design • Mechanical/ergonomic structure design	\$150,000 – \$300,000		Industrial design firms provide professional-grade appearance, ergonomics, material choices and manufacturability improvements.
	Cloud Backend, Big Data &	• Sleep report backend • Big data pipeline	\$300,000 – \$700,000		Critical for reports, data storage, security compliance and

Vendors & Outsourcing Costs	Security Development	• Subscription/payment backend • Security/privacy implementation		\$1,800,000 – \$4,000,000	subscription monetization.
	Mobile App Development (Partial Outsourcing)	• UI/UX design • IOS/Android front-end development • Bed control features integration	\$200,000 – \$500,000		Delivering seamless UX and integrating with cloud-outsourcing accelerates delivery.
	Third-Party Certification & Testing	• FCC, CE, UL certification • EMC testing • Lab Equipment rental	\$100,000 – \$300,000		Certification is mandatory for market entry since the smart bed is an electronic device with wireless and safety components.
	Outsourced Mass Production & Logistics	• Supply chain setup • Tooling & molds • Mass manufacturing • Packaging • Logistics & warehousing • After-sales system setup	\$1,000,000 – \$2,200,000		Manufacturing is fully outsourced in WBS 1.8 Production and Delivery, making this one of the largest vendor costs.
Technology (Cloud Infra, AI Compute & Tools)	Cloud Hosting & Storage	• Databases • Object storage • API hosting • Authentication	\$150,000 – \$300,000	\$400,000 – \$800,000	Data requires reliable, secure 24/7 cloud infrastructure.
	AI Training Compute	• GPU servers • Model training runtime • Dataset processing costs	\$100,000 – \$250,000		AI models (sleep staging, OSA detection) require intensive compute cycles.
	Software Tools	• CI/CD pipeline • Data labeling tools • API testing platforms • Developer tool	\$80,000 – \$150,000		Supports efficient model deployment, continuous integration and quality control.

		licenses			
	Monitoring & Post-Launch Observability Tools	- System performance dashboards - Error tracking tools - Uptime monitoring	\$70,000 – \$100,000		Supports WBS 1.9 Post-Launch Monitoring and ensures stable user experience.
Total Estimated Project Cost: \$4,500,000 – \$9,000,000					

2. Cost Breakdown by Project Phase

Phase	Cost Components	Estimated Cost
Phase 1 – Research & Concept (M1-2)	- Salaries (Product, PM, Architect) - Market research - Feasibility analysis - Early engineering consultation	\$150,000 – \$300,000
Phase 2 – Product & Technical Design (M3-4)	- Salaries (Product, Mechanical, System Architect) - Industrial design outsourcing - System architecture work - Early prototype material cost	\$800,000 – \$1,500,000
Phase 3 – Hardware + Software Development (M4-8)	- Salaries (Firmware engineers, AI scientists, Cloud backend, Mobile developer) - Sensor modules & PCB fabrication - Cloud infra costs - Algorithm computation cost (GPU training)	\$1,500,000 – \$3,000,000
Phase 4 – Testing + Certification (M8-11)	- Salaries (QA team, algorithm team for validation, PM) - EMC / safety certification fees - Medical-grade validation - Beta test program - Trial units material	\$800,000 – \$1,500,000
Phase 5 – Production & Launch (M11-12+)	- Salaries (Operations, PMO, Supply chain) - Outsourced mass production	\$1,200,000 – \$2,500,000

	• Packaging + logistics • Marketing launch campaigns • After-sales support setup	
--	--	--

3. Cost Control Approach

- Cost baseline aligned with Gantt Schedule
- Monthly cost review (WBS 1.10 Project Control)
- Vendor contracts with fixed-price structure to reduce overruns
- Contingency reserve: 10–20% due to outsourced manufacturing uncertainties

4. Key Cost Risks

Risk	Impact	Mitigation
Sensor/component failure requiring redesign	High	• Early validation testing • Industrial grade components
Certification failure requiring retesting	High	• Pre-compliance internal testing • Regulatory consultant involvement from Phase 2
Firmware/AI late-stage rework increasing engineering cost	High	• Automated regression tests • Code freeze before certification
Vendor manufacturing delays	Medium	• Dual-supplier strategy • Agreement with penalty clauses • Buffer inventory
Cloud service & backend cost overrun	Medium	• Usage monitoring • Architecture optimization • Periodic cost audits

Schedule Management (Ya cheng Hsieh)

Phase 1: Research & Concept (Month 1–2)

Project Proposal & Plan (M1–M2)

- Market Needs & Analysis
- Competitive analysis (Week 1–2)
- Target user persona research (Week 2–4)

Feasibility plan

- Technical feasibility (Week 3–5)
- Cost & ROI estimation (Week 4–5)
- Risk identification (Week 5)
- Regulatory feasibility check (Week 5–6)

Project Charter & Plan

- Project charter (Week 6)
- Preliminary project plan (Week 6–7)

Phase 2: Product & Technical Design (Month 3–4)

Product Design (M3–M4)

- Core function definition (Week 9–10)
- Sleep report logic & personalization (Week 10–12)

Appearance & Structural Design (M3–M4)

- Industrial design (Week 10–12)
- Mechanical/structural design (Week 11–13)

Technical Architecture (M3–M4)

- System architecture (Week 9–11)
- Sensor fusion scheme (Week 11–12)

AI Testing

- Add early prototype testing, unit tests, and sensor calibration validation while hardware and software are still under development. (Week 13-14)

Phase 3: Hardware & Software Development (Month 4–8)

Hardware Development (M4–7)

- Sensor selection (Week 13–15)
- Sensor layout & wiring design (Week 15–18)
- PCB + power + comms module development (Week 16–20)

Embedded & Algorithm Dev (M4–8)

- Firmware drivers & sync logic (Week 16–20)
- Edge processing (Week 18–22)
- AI model development:
 - Data fusion (Week 20–23)

- Sleep staging & posture detection (Week 23–28)
- Sleep scoring models (Week 26–30)
- Disorder screening models (Week 26–32)

Early Metric & AI Testing (M7)

- Early Metric Accuracy Testing (on prototype data) (W27–W30)
- Run AI model validation and data accuracy tests concurrently with model training.(W27-W30)

Cloud & Mobile App (M4–8)

- Mobile UI/UX design (Outsources) - runs in parallel
- Cloud API, storage, security (Week 20–27)
- Mobile App Development & Integration (OUTSOURCED) (W23–W31)
- Subscription backend (Week 26–30)
- Sleep report + bed control functions (Week 23–32)

Phase 4: Testing, Certification & Pilot Production (Month 8–11)

Product Testing (M8–11)

- Hardware stress & EMC testing (Week 33–36)
- Software & app testing (Week 34–36)
- AI accuracy testing with medical equipment (Week 36–40)
- System integration testing (Week 40–43)

Certification (M10–11)

- FCC/CE/UL certification (Week 40–48)

User Acceptance Testing (M10–11)

- Small batch prototypes (Week 40–44)
- Internal/external test users (Week 42–46)
- Feedback iteration (Week 46–48)

Phase 5: Mass Production & Launch (Month 11–12)

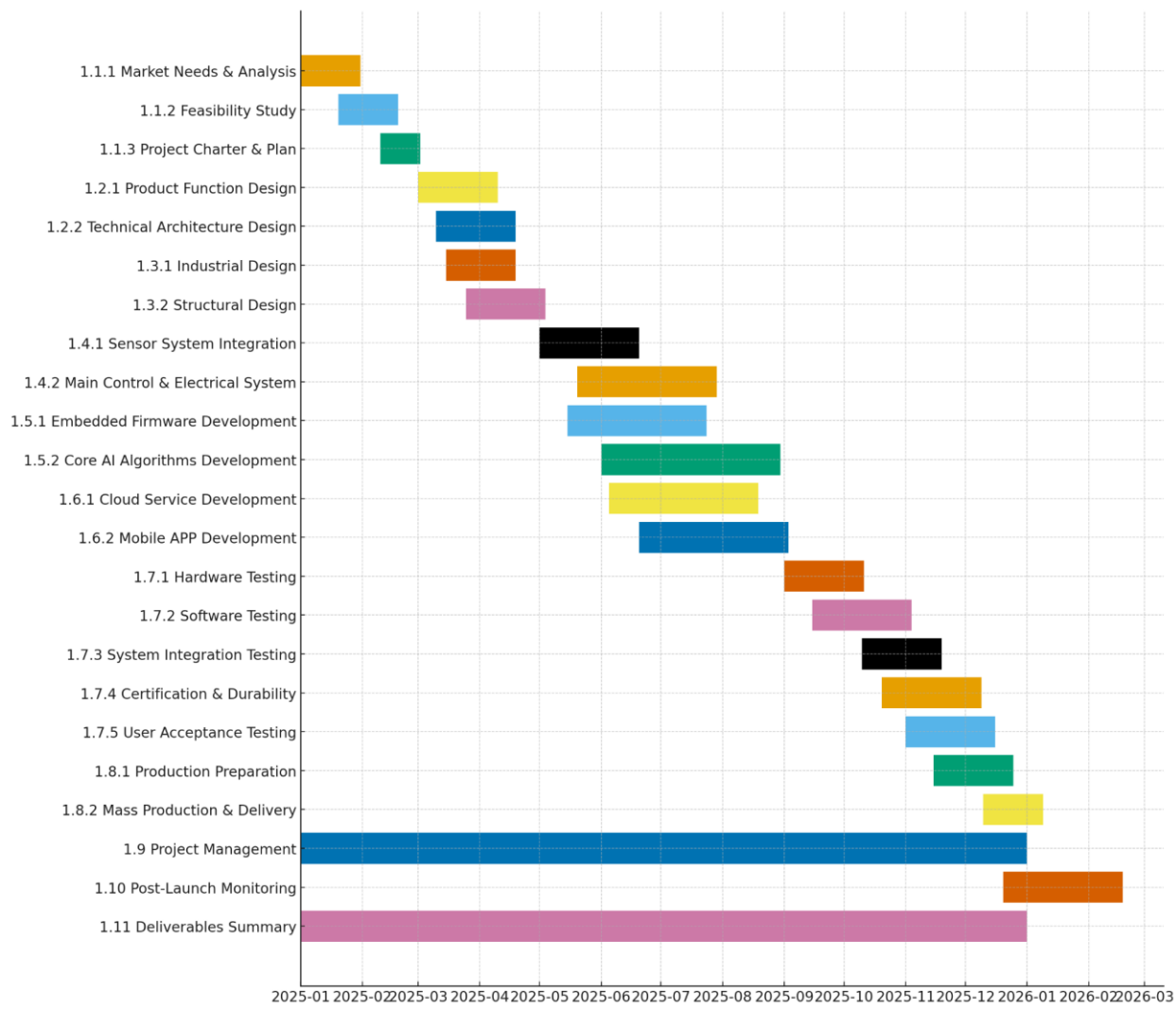
Production & Delivery

- Supply chain & production line setup (Week 44–48)
- Pilot production run troubleshooting (Week 45–48)
- Mass production (Week 48–52)
- Packaging & logistics build-out (Week 50–52)

Post-Launch Monitoring (M12+)

- User feedback monitoring
- App update plan
- System reliability analysis

G-natt Chart:



Communication Management (Ya cheng Hsieh)

Department	Communication Frequency	Purpose
Hardware Engineering	Daily / Weekly	Sensor selection, wiring, PCB development
Embedded Firmware	Daily	Driver stability, data packet consistency
AI Algorithm Team	2–3 times a week	Model performance, data alignment
Cloud / Backend	2–3 times a week	API, storage, analytics pipeline
Mobile App	Weekly	UI changes, backend integration
System Architecture	Bi-weekly	Ensure architecture stability
Product Management	Weekly	Feature alignment, requirements tracking
Industrial Design	Weekly	External appearance and structural design
Supply Chain	Weekly	Materials, delivery time, procurement risks
QA & Testing	Weekly	Test plans, bugs, verification results
Customer Support	Monthly	Launch preparation and issue tracking
Pilot Testers	Weekly	User feedback, sleep data validation
Sponsor / Leadership	Monthly	Strategic direction and escalation

How teams share updates:

Effective communication across the Smart Bed project relies on structured, predictable update mechanisms. Each team is required to share progress, risks, and dependency changes using standardized tools and formats to ensure transparency and alignment.

Daily & Weekly Update Mechanisms

- **Daily Stand-Up Meetings:**

Engineering teams such as hardware, firmware, AI, cloud, app share what they

worked on yesterday, plans for today, and anything else that requiring cross-team support.

- **Weekly Cross-Functional Syncs:**

Product, design, engineering, and supply chain teams meet weekly to review milestones, design changes, dependency shifts, and integration progress.

- **Reviews (Bi-weekly):**

All teams demonstrate completed work, highlight technical challenges, and gather feedback from stakeholders and leadership.

- **Jira Updates:**

All tasks, test results, API changes, and architectural diagrams must be documented in combine and tracked in Jira. Requirements and design changes must be logged immediately to avoid misalignment.

- **Status Dashboards:**

Engineering and product teams maintain real-time dashboards showing progress, open issues, risk levels, and readiness state.

How emergency signals and critical data will be communicated:

The Smart Bed will have a number of components to ensure safety and reliability. When emergency events occur such as sensor failure, firmware corruption, overheating, cloud downtime, or malfunctioning motors, communication must follow a strict escalation path with automated and manual actions.

1. Real-Time Device Alerts

- The firmware will automatically send emergency status codes to the cloud when abnormal data patterns, overheating, or malfunction are detected.
- The cloud backend immediately pushes alerts to:
 - Mobile App team
 - QA team
 - Cloud on-call engineer

This ensures rapidly solving the problem.

2. Automated Logging & Traceability

- All emergency incidents automatically generate logs stored in the cloud.
- Firmware crash logs, sensor logs, and AI inference logs are uploaded for post-analysis.

3. Communication to Non-Technical Teams

- Customer support receives a simplified emergency brief written by PM + engineering leads.

- Marketing and leadership are notified only if the emergency affects user safety or launch schedule.
- Compliance is looped in if the emergency involves safety tests or regulatory risks.

These actions can ensure user safety and system reliability, protect brand trust, and reduce the chance of repeated failures through structured analysis.

Risk Management (Kanokkan)

1. Technical Risks

- **Sensor Inaccuracy or Instability**

Sensors such as pressure pads, bio-radar, or temperature modules may perform differently once embedded inside the mattress, especially under varying weight and sleeping positions. If the readings become inconsistent, the sleep analysis and health metrics will be unreliable. This risk affects user trust and must be controlled through early calibration and prototype testing.

- **Firmware Integration Failures**

The data pipeline from sensors → MCU → cloud → mobile app is complex, and synchronization issues could cause data loss or delays. If the firmware cannot reliably transmit or preprocess data, users may not receive accurate sleep reports. Continuous integration testing is necessary to detect failures early.

- **AI Model Accuracy Below Requirements**

The sleep staging and anomaly detection algorithms must reach at least 85% accuracy, but poor training data or model overfitting can reduce performance. Inaccurate AI outputs can mislead users about their sleep health. Rigorous validation using medical equipment is required to mitigate this risk.

- **System Latency or Cloud Data Loss**

High server load, unstable Wi-Fi, or poor cloud architecture may delay sleep report generation or cause incomplete data. This directly impacts user experience and system reliability. Implementing buffering and redundancy helps reduce this risk.

2. Quality & Certification Risks

- **Insufficient Testing Coverage**

If hardware, firmware, cloud, and app components are not tested under all realistic scenarios (different weights, movements, temperatures), critical defects may go undetected. Limited test coverage increases the chance of failures during real user operation. A comprehensive test matrix ensures full coverage.

- **Failure to Pass FCC/CE/UL Certification**

The smart bed includes electronics and wireless components that must meet strict safety and electromagnetic standards. Failing certification can cause delays, redesign costs, and inability to launch. Running pre-compliance tests early reduces the risk of late surprises.

- **Late Discovery of High-Severity Defects**

If integration issues or major bugs are found too late in the schedule, the cost of fixing them increases significantly. Late-stage issues also disrupt the launch timeline. Early prototyping and regression tests help prevent this.

3. Data Security & Privacy Risks

- **Exposure of Sensitive Health Data**

Sleep patterns, heart rate, and breathing data are highly personal, and any cloud breach would have serious consequences. A data leak damages user trust and may violate privacy regulations. End-to-end encryption and strict access control mitigate this risk.

- **Cloud Server Downtime Affecting Report Generation**

If cloud services fail or experience latency, users may not receive their sleep reports on time. This degrades reliability and user satisfaction. Redundant server systems and monitoring tools help maintain uptime.

4. Operational & Supply Chain Risks

- **Shortage or Delay of Critical Sensors**

Key components, especially high-precision sensors or MCUs, may have long lead times or global shortages. Delays in procurement push back prototype development and testing schedules. Dual sourcing and early ordering help reduce supply risk.

- **Manufacturing Partner Delays**

Outsourced production may deliver parts late or with quality issues, disrupting the PVT and mass production stages. Miscommunication between design and manufacturing increases this risk. Frequent vendor check-ins and clear specification documents are required.

5. User Experience Risks

- **Mobile App Confusion or Instability**

If users struggle to navigate the app or encounter bugs, they may perceive the entire system as unreliable. Poor UI/UX can reduce adoption and satisfaction. Early usability testing ensures that the app is intuitive and stable.

- **Incorrect or Unreliable Emergency Alerts**

False alarms for apnea or abnormal heart rate can cause stress and decrease trust, while missed alarms can pose safety risks. Accurate thresholds and multi-sensor validation are required to ensure alert reliability.

6. Ongoing Risk Monitoring & Control

- **Continuous Review of Risks Across All Project Phases**

Risks are monitored weekly, and the Risk Register is updated as new issues appear or existing ones change in severity. Milestone-based quality gates ensure that no phase progresses without addressing critical risks. This continuous oversight helps the project maintain product safety, stability, and performance throughout development.

Commentary

1- How you organized your group project?

At the beginning of our group project, we set out to clearly define roles, responsibilities, quality and risk control, resource management, and communication protocols. Each member of the team chose one of the sections based on their individual strengths. After identifying each of our respective sections, we set up a collaborative workspace in which members of the team could upload drafts of documents and comment on each other's work. Finally, we established internal deadlines to make sure our team members completed their assigned tasks in a timely manner and had enough time to review and edit the team's work prior to submission in accordance with the project schedule.

2- How you proceeded in your collaboration efforts

The most important factors that contributed to our successful working partnership were continual open communication with a structured process for working together. These communication also provided an opportunity for us to review our written documents, ensure that each of our written sections were aligned in terms of tone and content, and make sure we achieve professor's requirement.

In addition to using the regular check-in meetings to track progress, we also utilized shared electronic documents so both teams could collaborate on writing and editing the work. Moreover, when shared decisions were made that impacted more than one written section, both teams would work collaboratively to ensure that all written sections were aligned with one another, thereby minimizing the impact of any divergent written content due to poor communication.

3- How you decided to finalize the approach or focus for the final draft?

To finalize the approach, we compared all sections to ensure they matched the overall project objectives and followed the same structure as the professor's expectations. We also standardized formatting, wording, and examples so the document read as a single unified report. After consolidating all parts, we conducted one final group review to confirm accuracy, fill gaps, and polish the final version.

4- What were the challenges you faced as a team?

One challenge we faced was aligning everyone's writing style and working pace, since each member interpreted the assignment slightly differently at first. Moreover, because everyone is busy with their own schedules every day, finding a time when everyone can discuss the report was a big challenge for us throughout the entire process. However, through discussion, shared editing, and feedback iterations, we were able to overcome these challenges and produce a coherent and well-structured final report.

5- what was good about the experience.

Working on this group project was a highly positive and meaningful experience because it allowed our team to collaborate across different functional perspectives, just like a real product development environment. Each member contributed unique strengths—some focused on technical understanding such as firmware, AI, or system architecture, while others concentrated on planning, scheduling, and communication management. Through our structured communication process, such as daily updates, weekly syncs, and continuous progress tracking, we were able to stay aligned, divide responsibilities clearly, and avoid misunderstandings. The experience also strengthened our ability to break down a complex project into phases, assign roles, and integrate different

components into a cohesive schedule. Overall, the project improved our teamwork skills, enhanced our understanding of cross-functional collaboration, and helped us practice the kind of communication and coordination required in real-world product development.